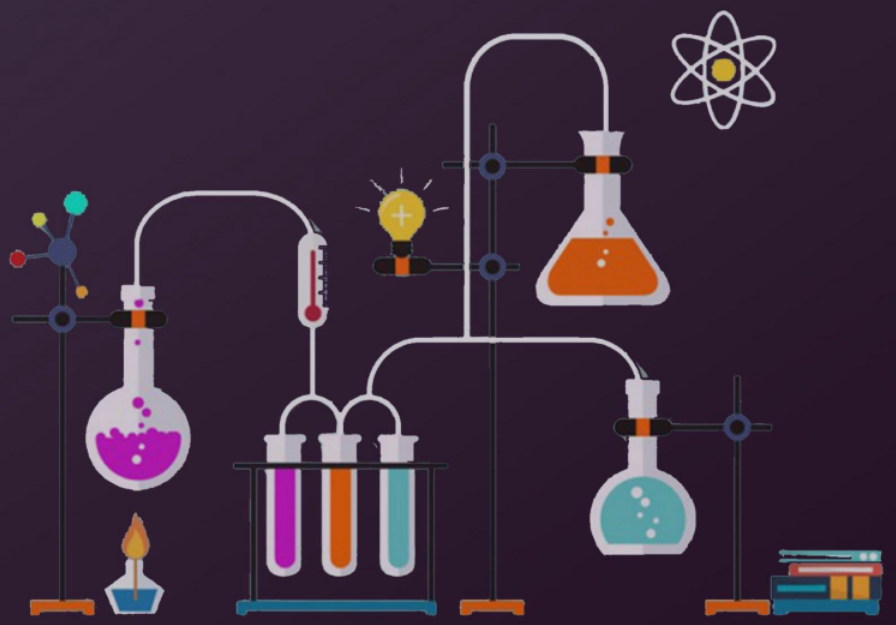
CHEMISTRY

Thermochemistry



SHEET

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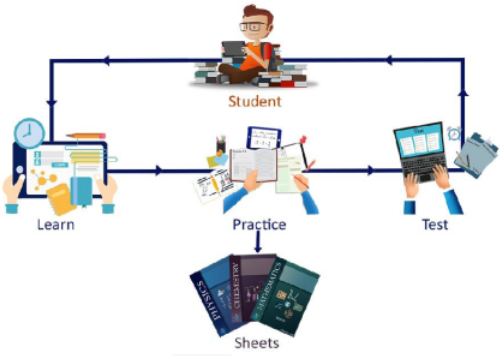
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 4
6.	Exercise 2A	Page - 7
7.	Answer Key	Page - 9

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Basic Definitions	1-6	1,7	1-8	1,8
2	Calorimetry, Bond Enthalpy, Enthalpy of Resonance and Neutralisation	7-13	2-5	9-12	2-4
3	Enthalpy of Solution, Hydration Enthalpy and Bond Haber Cycle	14-18	6,8	13-15	5,10-13
4	ACC	19-22	9	16-19	6,7,9

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Basic Definitions

Q1. For which of the following, standard formation enthalpy is zero :-

(A) $Br_2(\ell)$ (B) $C_{(graphite)}$

(C) (1) & (2) both (D) None of these

Q2. Hess law is applicable for the determination of heat of?

(A) Reaction (B) Transition

(C) Formation (D) All of these

Q3. Calorific value of H_2 gas is x kJ/gm what is heat of formation of H_2O :-

(A) x kJ (B) 2x kJ

(C) $\frac{x}{2}$ kJ (D) 18 x kJ

Q4. Enthalpy of formation of OH^-(aq.) (in kcal/mol) will be **?** ($\Delta_f H^{o+} = 0$)

Given : ΔH°_f (H_2O, l) = –68 kcal/mol

(A) 34 (B) –81.7

(C) –54.3 (D) 47.7

Q5. Given that $C + O_2 \longrightarrow CO_2$, $\Delta H^0 = -x$ KJ, $2CO + O_2 \longrightarrow 2CO_2$; $\Delta H^0 = -y$ KJ The enthalpy of formation of carbon mono-xide will be ?

(A) y – 2x (B) $\frac{2x-y}{2}$

(C) $\frac{y-2x}{2}$ (D) 2x – y

Q6. The heat of formation of two compounds A and B are –84 kJ and –156 kJ respectively. Which one of the following statement is correct ?

(A) A is more stable than B

(B) A is less stable than B

(C) Both A and B are unstable

(D) A and B are equally stable

Calorimetry, Bond Enthalpy, Enthalpy of Resonance and Neutralisation

Q7. For the reaction $CH_3COOH(l) + 2O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(l)$ at 25°C and 1 atm. pressure, ΔH = –874 kJ. Then the change in internal energy is (ΔU) is .....

(A) – 874 kJ (B) – 871.53 kJ

(C) – 876.47 kJ (D) + 874 kJ

Q8. Which of the following statement is correct:-

(A) ΔH is positive for exothermic reaction

(B) ΔH is negative for endothermic reaction

(C) The heat of neutralization of strong acid and strong base is always the same

(D) The enthalpy of fusion is negative

Q9. Bond dissociation enthalpy of H_2, Cl_2 and HCl are 434, 242 and 431 kJ mol^{-1} respectively. Enthalpy of formation of HCl is ?

(A) –93 kJ mol^{-1} (B) 245 kJ mol^{-1}

(C) 93 kJ mol^{-1} (D) –245 mol^{-1}

Q10. The molar heat of formation of NH_4NO_3(s) is–367.54 kJ and those of N_2O(g) and $H_2O(\ell)$ are + 81.46 kJ and –285.78 kJ respectively at 25°C and 1.0 atmospheric pressure. Calculate ΔU at 25°C for the reaction :

$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(\ell)$

(A) –122.56 kJ (B) –125.04 kJ

(C) –140.34 kJ (D) –112.04 kJ

Q11. What amount of energy (kJ) is released in the combustion of 5.8g of $C_4H_{10}(g)$

$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(\ell)$;

$\Delta H^\circ = -5756$ kJ :-

(A) 575.6 (B) 287.8

(C) 182 (D) 57.56

Q12. The enthalpy of neutralization of which of the following types of acids and bases is nearly same

(A) HCN and NaOH

(B) HCl and KOH

(C) HCN and KOH

(D) HCl and NH_4OH

Q13. Heat of neutralisation of the reaction :-

$NaOH + HCl \rightarrow NaCl + H_2O$

is -57.1 kJ mol^{-1}.

The heat released when 0.25 mole of NaOH is titrated against 0.25 mole of HCl will be?

(A) 22.5 kJ (B) 57.1 kJ

(C) 28.6 kJ (D) 14.3 kJ

Enthalpy of Solution, Hydration Enthalpy and Born Haber Cycle

Q14. Heat released upon combustion of 2gm of $C_{graphite}$ is X KJ. Heat of formation of CO_2 is?

(A) $-6X$ KJ (B) $-X$ KJ

(C) $+6X$ KJ (D) $+X$ KJ

Q15. The data below refers to gas phase reaction at constant pressure at 25°C :

$CH_3{-}CH_3 \rightarrow CH_3{-}\dot{C}H_2 + H$;

$\Delta H_1 = +420$ KJ mol^{-1}

$CH_3{-}\dot{C}H_2 \rightarrow CH_2{=}CH_2 + H$;

$\Delta H_2 = +168$KJ mol^{-1}

from these data, the enthalpy change ΔH for reaction,

$2CH_3{-}\dot{C}H_2 \longrightarrow CH_3{-}CH_3 + CH_2{=}CH_2$ is ?

(A) +250 KJ (B) +588 KJ

(C) -252 KJ (D) -588 KJ

Q16. The enthalpy change for the reaction :-

$2C_{(graphite)} + 3H_2(g) \rightarrow C_2H_6(g)$ is called :

(A) Enthalpy of formation

(B) Enthalpy of combustion

(C) Enthalpy of hydrogenation

(D) Enthaply of vaporisation

Q17. $\Delta H_{f(x)}$, $\Delta H_{f(y)}$, $\Delta H_{f(R)}$ and $\Delta H_{f(S)}$ denote the enthalpies of formation of x, y, R and S respectively. The enthalpy of the reaction, $x + y \rightarrow R + S$ is given by :-

(A) $\Delta H_{f(x)} + \Delta H_{f(y)}$

(B) $\Delta H_{f(R)} + \Delta H_{f(S)}$,

(C) $\Delta H_{f(x)} + \Delta H_{f(y)} - \Delta H_{f(R)} - \Delta H_{f(S)}$

(D) $\Delta H_{f(R)} + \Delta H_{f(S)} - \Delta H_{f(x)} - \Delta H_{f(y)}$,

Q18. Gibb's energy change for the dissolution of potassium nitrate in water at 298 K is –

$KNO_3(S) + water \longrightarrow K^+(aq) + NO_3^-(aq)$

$\Delta H^0 = 34$ KJ mol^{-1}

$\Delta S^0 = 0.116$ KJ K^{-1} mol^{-1}

(A) +0.568 KJ mol^{-1}

(B) -0.568 KJ mol^{-1}

(C) -1.036 KJ mol^{-1}

(D) +1.036 KJ mol^{-1}

ACC

Q19. Given the following equations and ΔH° values, determine the enthalpy of reaction at 298 K for the reaction :-

$C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$

$H_2(g) + F_2(g) \rightarrow 2HF(g)$; $\Delta H^\circ_1 = -537$ kJ

$C(s) + 2F_2(g) \rightarrow CF_4(g)$; $\Delta H^\circ_2 = -680$ kJ

$2C(s) + 2H_2(g) \rightarrow C_2H_4(g)$; $\Delta H^\circ_3 = 52$ kJ

(A) -1165 kJ

(B) -2486 kJ

(C) +1165 kJ

(D) +2486 kJ

Q20. The heats of combustion of carbon and carbon monoxide are –390 kJ and –278 kJ respectively. The heat of formation of carbon monoxide is ?

(A) –112 kJ (B) –668 kJ
(C) 112 kJ (D) 668 kJ

Q21. If enthalpies of methane and ethane are respectively 320 and 360 calories then the bond energy of C – C bond is ?

(A) 80 calories (B) 40 calories
(C) 60 calories (D) 120 calories

Q22. Na(s) $\longrightarrow$ Na(g), the heat of reaction is called as -

(A) Heat of vaporisation
(B) Heat of atomisation
(C) Heat of sublimation
(D) Both (B) and (C)

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Hess's law states -

(A) H = U + PV
(B) It is not possible to create or destroy energy
(C) Enthalpy change of a given chemical reaction is same whether the process occurs in one or more sequence of steps
(D) Total entropy of the universe remains constant

Q2. Which of following is/are correct statements?

(A) If $\Delta_r C_p = 0$ then enthalpy of reaction remains constant on increase in temperature.
(B) If $\Delta_r C_p$ is positive then enthalpy of reaction increase on increase in temperature.
(C) In endothermic reaction, enthalpy of reaction may increase when temperature increase.
(D) For a chemical reaction ΔH does not depend on temperature

Q3. Which of following reaction is/are combustion reaction -

(A) $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(\ell)$

(B) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

(C) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$

(D) $N_2(g) + O_2(g) \rightarrow 2NO(g)$

Q4. The enthalpy of combustion of a substance-

(A) Is always positive
(B) Is always negative
(C) Can be either zero or greater than zero
(D) Is unpredictable till calculations are done.

Q5. Under the same conditions, how many mL of 1M KOH & 0.5 M H_2SO_4 solutions, respectively when mixed for a total volume of 100 mL produces the highest rise in temperature.

(A) 67 : 33 (B) 33 : 67
(C) 40 : 60 (D) 50 : 50

Q6. $HA + OH^- \rightarrow H_2O + A^- + q_1$ kJ

$H^+ + OH^- \rightarrow H_2O + q_2$ kJ

The enthalpy of dissociation of HA is ?

(A) $(q_1 + q_2)$ (B) $(q_1 - q_2)$
(C) $(q_2 - q_1)$ (D) $-(q_1 + q_2)$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

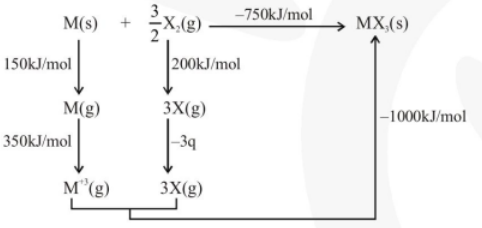
Q7.

Column-I	Column-II
(A) Endothermic reactions	(P) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$
(B) Exothermic reaction	(Q) $NH_3(g) \longrightarrow N(g) + 3H(g)$
(C) $\|\Delta H\| = \|\Delta U\|$	(R) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$
(D) $\|\Delta H\| \neq \|\Delta U\|$	(S) $\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow HCl(g)$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q8. Consider the following Born Haber cycle for the formation of $MX_3(S)$



Calculate the value of $\frac{2q}{100}$.

where q is in $kJmol^{-1}$

Q9. The Heat of formation of liquid octane at 300 K is – 55.6 K Cal mol^{-1} under constant volume. If Heat of formation under constant pressure is Z (Kcal mol^{-1}) than calculate the value of $\frac{Z}{61}$.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Heat of combustion of CH_4, C_2H_4, C_2H_6 are –890, –1411 and – 1560 kJ/mol respectively. Which has the lowest fuel value is ?

(A) CH_4 (B) C_2H_4

(C) C_2H_6 (D) All of these

Q2. The bond dissociation energies for single covalent bonds formed between carbon and A, B, C, D and E atoms are:-

Bond	Bond energy (kcal mol^{-1})
(i) C – A	240
(ii) C – B	382
(iii) C – D	276
(iv) C – E	486

This indicates that the smallest atom is ?

(A) A (B) B

(C) D (D) E

Q3. The ΔH_f° for $CO_2(g)$, $CO(g)$ and $H_2O(g)$ are -393.5; -110.5 and -241.8 kJ mol^{-1} respectively. The standard enthalpy change (in kJ) for the reaction

$CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$ is ?

(A) 524.1 (B) 41.2

(C) –262.5 (D) –41.2

Q4. Hess law of heat summation includes -

(A) Initial reactants only

(B) Initial reactants and final products

(C) Final products only

(D) Intermediates only

Q5. $H_2(g) + ½\ O_2(g) \rightarrow H_2O(g)$

B.E. (H–H) = x_1 ; B.E. (O = O) = x_2 ;

B.E. (O – H) = x_3

Heat of vaporisation of water = x_4 then ΔH_f (heat of formation of liquid water) is :-

(A) $x_1 + \frac{x_2}{2} - x_3 + x_4$

(B) $2x_3 - x_1 - \frac{x_2}{2} - x_4$

(C) $x_1 + \frac{x_2}{2} - 2x_3 - x_4$

(D) $x_1 + \frac{x_2}{2} - 2x_3 + x_4$

Q6. The standard heat of combustion of solid boron is equal to :-

(A) $\Delta H_f^\circ(B_2O_3)$ (B) $\frac{1}{2}\Delta H_f^\circ(B_2O_3)$

(C) $2\Delta H_f^\circ(B_2O_3)$ (D) $-\frac{1}{2}\Delta H_f^\circ(B_2O_3)$

Q7. When ethyne is passed through a red hot tube, then formation of benzene takes place. Calculate the standard heat of trimerisation of ethyne to benzene.

$$3C_2H_{2(g)} \longrightarrow C_6H_{6(g)}$$

Given :

$\left(\Delta H_f^\circ\right)_{C_2H_2(g)} = 230\,KJ\,mol^{-1}$,

$\left(\Delta H_f^\circ\right)_{C_6H_{6(g)}} = 85\,KJ\,mol^{-1}$

(A) –145 KJ/mole (B) 605 KJ/mole

(C) –605 KJ/mole (D) –405 KJ/mole

Q8. Which of the following reaction defines ΔH_f° :-

(A) $H(g) + F(g) \rightarrow HF(\ell)$

(B) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

(C) $C_{diamond} + O_2(g) \rightarrow CO_2(g)$

(D) $\frac{1}{2}H_2(g) + \frac{1}{2}F_2(g) \rightarrow HF(g)$

Q9. Values of heat of formation for SiO_2 and MgO are –48.4 and –34.7 kJ respectively. The heat of the reaction :

$2Mg + SiO_2 \rightarrow 2MgO + Si$ is ?

(A) 21.0 kJ (B) –21.0 kJ

(C) 13.7 kJ (D) –13.7 kJ

Q10. Compounds with high heat of formation are less stable because :-

(A) High temperature is required to synthesise them

(B) Molecules of such compounds are distorted

(C) It is difficult to synthesis them

(D) Energy rich state leads to instability

Q11. Given that -

(i) ΔH°_f of N_2O ($N \leftarrow N=O$) is 82 KJ mol^{-1}

(ii) Bond energies of N≡N, $N \leftarrow N$, O=O and N=O are 946, 418, 498 and 607 KJ mol^{-1} respectively.

The resonance energy of N_2O is ?

(A) –44KJ (B) –66 KJ

(C) –62 KJ (D) –88 KJ

Q12. Enthalpy of neutralisation of HCl with NaOH is x. The heat evolved when 500 ml of 2N HCl are mixed with 250 ml of 4 N NaOH will be ?

(A) 500 x (B) 100 x

(C) x (D) 10 x

Q13. In which of the following reactions does the heat change represent the heat of formation of water

(A) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$;
$\Delta H = -116$ kcal

(B) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$;
$\Delta H = -58$ kcal

(C) $H^+(aq.) + OH^-(g) \rightarrow 2H_2O(l)$;
$\Delta H = -116$ kcal

(D) $C_2H_2(g) + 2\frac{1}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(l)$;
$\Delta H = -310$ kcal

Q14. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$; $\Delta H = -68.39$ kcal

$K + H_2O + \text{Water} \rightarrow KOH(aq) + \frac{1}{2}H_2$;

$\Delta H = -48$ kcal

$KOH + \text{Water} \rightarrow KOH(aq)$; $\Delta H = -14$ kcal

The heat of formation of KOH (in kcal) is:-

(A) $-68.39 + 48 - 14$

(B) $-68.39 - 48 + 14$

(C) $68.39 - 48 + 14$

(D) $68.39 + 48 + 14$

Q15. If $\Delta_f H°$ (C_2H_4) and $\Delta_f H°$ (C_2H_6) are x_1 and x_2 kcal mol^{-1}, then heat of hydrogenation of C_2H_4 is ?

(A) $x_1 + x_2$ (B) $x_1 - x_2$

(C) $x_2 - x_1$ (D) $x_1 + 2x_2$

Q16. If ΔH is the change in enthalpy and ΔU the change in internal energy accompanying a gaseous reaction

(A) ΔH is always greater than ΔU

(B) $\Delta H < \Delta U$ only if the number of moles of the products is greater than the number of the reactants

(C) ΔH is always less than ΔU

(D) $\Delta H < \Delta U$ only if the number of moles of the products is less than the number of moles of the reactants

Q17. Given that

$2C(s) + 2O_2(g) \rightarrow 2CO_2(g)$; $\Delta H = -787$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$; $\Delta H = -286$ kJ

$C_2H_2(g) + 2\frac{1}{2}O_2(g) \rightarrow 2CO_2(g) + H_2O(\ell)$;

$\Delta H = -1301$ kJ

Heat of formation of acetylene is ?

(A) –1802 kJ \

(B) +1802 kJ

(C) –800 kJ

(D) +228 kJ

Q18. Heats of combustion for $C_{(s)}$, $H_{2(g)}$ and $CH_{4(g)}$ are –394.8, –285.6 and –894.6 kJ mol^{-1} respectively. The value of ΔH for the reaction: $C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)}$ is ?

(A) –52 kJ (B) –63 kJ

(C) –71.4 kJ (D) –170 kJ

Q19. The bond energy of C = C and C – C at 298 K are 590 and 331 kJ mol^{-1} respectively.

The enthalpy of polymerisation per mole of ethylene is ?

(A) –70 kJ (B) – 72 kJ

(C) + 72 kJ (D) –68 kJ

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The net heat change in a chemical reaction is same whether it is brought about in two or more different ways in one or several steps. It is known as -

(A) Hess's law

(B) Law of conversion of energy

(C) Henry's law

(D) Joule's principle

Q2. Which is/are correct statements?

(A) Enthalpy of reaction of conversion of one entantiomer to another enantiomer is zero.

(B) Enthalpy of reaction of conversion of one isomer to another isomer is zero.

(C) Enthalpy of precipitation reaction is not equal to zero.

(D) Enthalpy of hydration is always negative

Q3. Which is/are correct statements?

(A) The energy released in neutralisation of HF and NaOH is greater than 57.3 $kJmol^{-1}$

(B) $HF\,(aq) \rightarrow H^+\,(aq) + F^-\,(aq)$ is exothermic reaction.

(C) $HCN\,(aq) \rightarrow H^+(aq) + CN^-\,(aq)$ is exothermic reaction.

(D) $HCl\,(aq) \rightarrow H^+(aq) + Cl^-\,(aq)$ $\Delta_r H = 0$

Q4. The Bond-energies of C ≡ C, C—H, H—H and C = C are 198, 98, 103, 145 kcal respectively. The enthalpy change of the reaction $HC \equiv CH + H_2 \rightarrow C_2H_4$ is ?

(A) – 152 kcal (B) 96 kcal

(C) 48 kcal (D) 80 kcal

Q5. From the following data make the correct options :-

Given :

$H^+(aq) + OH^-(aq) \longrightarrow H_2O(\ell) + 57.3$ kJ

$\Delta H_{solution}$ of HA (g) = –70.7 kJ/mol

$\Delta H_{solution}$ of BOH(g) = –20 kJ/mol

$\Delta H_{ionisation}$ of HA = 15 kJ/mol

$\Delta H_{ionisation}$ of BOH = 0

Reaction :-

(A) $HA(aq) + BOH(aq) \rightarrow BA(aq.) + H_2O(\ell)$; $\Delta_r H = -42.3$

(B) $HA(g) + BOH(g) \rightarrow BA(aq.) + H_2O(\ell)$; $\Delta_r H = -93$

(C) $HA(g) \rightarrow H^+(aq.) + A^-(aq)$; $\Delta_r H = -55.7$

(D) $B^+(aq) + OH^-(aq) \rightarrow BOH(aq.)$; $\Delta_r H = -20$

Q6. Calculate the energy required to excite one litre of hydrogen gas at 1 atm and 298 K to the first excited state of atomic hydrogen. The energy for the dissociation of H–H bond is 436 $kJ\ mol^{-1}$.

(A) 100 kJ (B) 18.65 kJ

(C) 95.02 kJ (D) 85.82 kJ

Q7. Which is incorrect statement?

(A) $IE_1 < IE_2 < IE_3$

(B) Energy absorbed when 2nd electron added

(C) $IE_3 < IE_2 < IE_1$

(D) $IE_2(Mg) > IE_1(Mg)$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8.

Column I	Column II
(A) Endothermic reaction with $\Delta_r S > 0$	(P) spontaneous reaction at any temperature
(B) Endothermic reaction with $\Delta_r S < 0$	(Q) spontaneous reaction at lower temperature
(C) Exothermic reaction with $\Delta_r S > 0$	(R) At higher temperature spontaneous reaction
(D) Exothermic reaction with $\Delta_r S < 0$	(S) Non spontaneous reaction at any temperature

Q9.

Column I	Column II
(A) $Ca^{+2}(aq) + CO_3^{-2}(aq) \longrightarrow CaCO_3(s)$	(P) $\Delta H_{formation}$
(B) $H_2O(\ell) \longrightarrow 2H(g) + O(g)$	(Q) $\Delta H_{isomerisation}$
(C) $C_2H_5OH(\ell) \longrightarrow CH_3OCH_3(\ell)$	(R) $\Delta H_{Precipitation}$
(D) $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \longrightarrow NH_3(\ell)$	(S) $\Delta H_{atomisation}$

TRIPLE MATCH

	Column-I	Column-II	Column-III
(i)	Enthalpy of neutralisation	(a) $2Sn(White) + O_2(g) \rightarrow 2SnO(s)$	(p) $\Delta_r H = +ve$
(ii)	Enthalpy of combustion	(b) $HClO_4(aq) + KOH(aq) \rightarrow KClO_4(aq) + H_2O(\ell)$	(q) $\Delta_r H = -ve$
(iii)	Enthalpy of formation	(c) $C(diamond) + O_2(g) \rightarrow CO_2(g)$	(r) $\Delta_{surrounding} S = -ve$
(iv)	Enthalpy of solution	(d) $CuSO_4 \cdot 5H_2O(s) + (aq) \rightarrow CuSO_4(aq)$	(s) $\Delta G = -ve$ at any temp.

Q10. Which is/are correct statements?

(A) (i)-(b)-(s) (B) (ii)-(a)-(s)

(C) (ii)-(c)-(q) (D) (iv)-(d)-(r)

Q11. Which is/are correct statements?

(A) (i)-(b) (B) (ii)-(c)

(C) (iii)-(a) (D) (iv)-(c)

Q12. Which is/are correct statements?

(A) (a)-(s) (B) (b)-(q)

(C) (d)-(r) (D) (c)-(p)

Q13. Which is/are correct statements?

(A) (i)-(s) (B) (ii)-(r)

(C) (iii)-(a) (D) (iv)-(b)

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	D	B	C	C	B	A	C	A	B
11	12	13	14	15	16	17	18	19	20
B	B	D	A	C	A	D	B	B	A
21	22								
D	D								

Ex-1A

1. C **2.** A,B,C **3.** A,B,C **4.** B **6.** D

5. C **7.** A→Q ; B→P,R,S ; C→S ; D→P,Q,R **8.** 3

9. 1

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	D	B	B	C	B	C	D	B	D
11	12	13	14	15	16	17	18	19	
D	C	B	B	C	D	D	C	B	

Ex-2A

1. A **2.** A,C **3.** A,B,D **4.** D **5.** A, C

6. B **7.** C

8. A→R; B→S; C→P; D→Q **9.** A→R; B→S; C→Q; D→P

10. C **11.** A,B **12.** B,C **13.** A

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